

KMT-2026-BLG-0521 / OGLE-2026-BLG-0058

The Gaia-pinned reduction (v4) — standalone report

Prepared 2026-09-07. Solution: `~/KMTdata/Results/v4b/IFfinal_<field>.mat` (variable IFsys). This document describes **only** this run. The decade reduction that remains the project's reference solution is described in `KMT260058_astrometry_report.pdf`.

1. What this run is

A reduction built to a different specification from the reference one, to test two ideas: restrict the analysis to a clean calibration set, and tie the frame to Gaia from the start rather than afterwards.

1. **Analysis set.** $14 < I_{\text{OGLE}} < 17$, no companion brighter than $I = 18$ within 2.0 arcsec, and a Gaia counterpart with **RUWE < 1.4**. The target is added explicitly — at $I = 18.13$ it falls outside the magnitude window. Every other source is **dropped at the input**, so all measurements, detrending and quality cuts act on this set alone. 2. **The frame is pinned to Gaia.** Each calibration star's position *and* proper motion are held at Gaia's values, converted into cut-out pixels. The per-epoch affine is therefore fitted against Gaia positions propagated to that epoch by Gaia's own motions. The target is left free; pinning it would erase the signal. The six-parameter gauge is gone by construction and no post-hoc tie is needed. 3. **Passengers.** Three stars at the target's own magnitude ($I \sim 18$), outside the calibration set, are carried along unpinned so that something comparable to the target can be plotted in the same frame.

	BLG41	BLG01
sources in the input MSc	1177	1187
analysis set	130	100
of which pinned to Gaia	126	96
free: target + passengers	4	4
epochs	19 133	19 981

2. The calibration set

Selection, all conditions required:

- $14 < I_{\text{OGLE}} < 17$
- **no companion brighter than $I = 18$ within 2.0 arcsec** (5 KMT pixels at 0.4"/pix), the companion list being the OGLE catalogue, which resolves about six times more sources than KMT detects
- **a Gaia DR3 counterpart with RUWE < 1.4**

	BLG41	BLG01
in the magnitude window, isolated	144	127
with a Gaia counterpart	144	116

The RUWE cut removes 12% and 17% — consistent with the 87% pass rate for $G < 17$ across the whole cone, so crowding is not biasing it heavily.

3. Gaia

RUWE is not in the local catsHTM DR3 table. It carries `astrometric_chi2_al` and `astrometric_n_good_obs_al` but neither RUWE nor the `u0(G, BP-RP)` table needed to normalise UWE into it. The true values were queried from the Gaia archive instead — 9464 sources in a 3 arcmin cone, cached in `~/KMTdata/GaiaRef/`. This mattered: at $G < 17$, **87%** pass RUWE < 1.4, where the unnormalised UWE proxy predicted 74%, so the proxy would have discarded about 13% of good stars for no reason.

Gaia's proper-motion errors for these stars are **0.070 / 0.044 mas/yr**, some fifty times better than our internal frame. That is what makes pinning worthwhile in principle.

The Gaia-to-pixel map is fitted directly from the matched pairs, not carried through OGLE. The OGLE bridge is needed only to identify which star is which; routing coordinates through it inherits OGLE's ~110 mas astrometric error.

	through OGLE	fitted directly
BLG41, median offset / scatter	252 / 159 mas	23 / 16 mas
BLG01	351 / 148 mas	21 / 20 mas

The fitted scale is 2.519 (BLG41) and 2.529 (BLG01) pix/arcsec against the 2.500 expected at 0.4"/pix — an independent check that the map is right. The remaining ~20 mas is static per star, so it displaces each star's residuals by a constant and does not affect proper motion or any time-dependent signal.

4. The chain

With proper motions supplied by Gaia, the joint proper-motion fit and the per-season fits of the reference reduction are unnecessary. Two steps remain:

1. **Full-decade SysRem correction**, frame pinned, 2 components on the astrometric residuals. 2. **Final solution** reusing that correction, with the pixel-phase term on, 15 weighted iterations before SysRem and 6 after.

Convergence over the frame stars, last three iterations: **0.0009 mas (BLG41)** and **0.0026 mas (BLG01)**.

5. Results

	BLG41	BLG01
frame-star residual RMS, $\Delta X / \Delta Y$	6.13 / 5.20 mas	5.30 / 5.42 mas
target residual RMS, $\Delta X / \Delta Y$	29.62 / 22.63 mas	28.73 / 23.25 mas

For comparison, the reference reduction gives bright-star 6.41 / 6.88 and 6.48 / 7.01, and target **24.62 / 21.00** and **25.48 / 21.78**.

The calibration stars are measured better and the target worse. The stars are bright and isolated, which is what

they were selected for; the target is **14 to 20% worse** than in the reference reduction. The cause is the frame: it is now built from 126 stars rather than ~600 weighted sources, which roughly doubles its ideal noise and raises its leverage by a factor of twenty to fifty. Restricting the analysis set is what costs. The Gaia tie is not at fault.

6. Proper motions, and whether the two fields agree

Converting the target's motion out of pixel axes with each field's own map:

	BLG41	BLG01	difference
v4, Gaia-pinned	-1.799 / -6.780	-1.158 / -7.147	0.640 / 0.367
reference, post-hoc tie	-2.92 / -7.25	-2.22 / -7.24	0.702 / 0.004

mas/yr, as $(\mu_\alpha \cos \delta, \mu_\delta)$. Both agree at about the 1 mas/yr level, which is the target's own uncertainty, but **pinning the frame did not tighten the agreement**. The post-hoc tie was already doing this job.

The passengers agree better than the target does. Three stars at the target's magnitude, freely fitted, carried through both fields:

star	BLG41	BLG01	difference
l = 18.14, 4.5 pix from the target	-0.63 / -5.47	-1.08 / -5.47	0.45 / 0.00
l = 17.91, 21 pix	-6.81 / -9.99	-6.94 / -10.06	0.13 / 0.07
l = 18.05, 48 pix	+3.06 / -6.34	+2.80 / -6.48	0.26 / 0.14
the target, l = 18.13	-1.80 / -6.78	-1.16 / -7.15	0.64 / 0.37

The two clean passengers reproduce two to five times better than the target. The 4.5-pix star's apparent agreement is not a good sign — see section 7.

Sensitivity to the set. Adding only those three passengers to a set of 127 moved the fitted target motion by **0.42 / 0.63 mas/yr on BLG41** and 0.09 / 0.41 on BLG01. A result that moves by half a mas per year when three faint stars join the set is not a precise result, and this is the most direct measurement here of how uncertain the target's motion actually is.

7. A blended neighbour tracks the event

The passenger 4.5 pixels from the target behaves unlike anything else in the run. Comparing the 2026 season against the 2016–2022 baseline:

star	distance	ΔX (BLG01 / BLG41)	ΔY (BLG01 / BLG41)	season scatter
target	0	+2.7 / +2.6	+7.1 / +6.4	3.6–5.3
passenger 1	4.5 pix	-44.1 / -57.8	-27.9 / -32.2	16–19
passenger 2	21 pix	+5.6 / +3.5	+2.9 / +3.3	2.3–3.7
passenger 3	48 pix	-3.4 / -5.5	-2.5 / -2.1	2.1–5.3

mas. The excursion is **40 to 58 mas at 4.5 pixels and gone by 21 pixels**, reproduced independently in both fields, and that star's season-to-season scatter is four times the others' in every season, not only in 2026. With a seeing FWHM

near 7 pixels, a star 4.5 pixels away shares most of its light with the target: as the target brightens through the event, the blend's centroid moves. The neighbour moves *away* from the target, consistently in both fields, which is what PSF fitting a blend tends to do as flux is reattributed to the brightening component.

Three consequences:

- **That star is unusable as a comparison.** It was chosen as the closest at the target's magnitude, and being closest is exactly what disqualifies it. Passengers 2 and 3 are the meaningful ones.
- **Its good inter-field agreement in section 6 is misleading.** Both fields see the same blending artefact from the same event, so agreeing is not evidence of a good measurement.
- **It bears directly on the target.** The blend is mutual. If the neighbour's centroid moves 40 to 58 mas when the target brightens, the target's centroid is being pulled by the same shared flux. The target's own +2.7 / +7.1 mas in 2026 therefore cannot be read cleanly as astrometric microlensing; some or all of it may be blending. That is a systematic at the tens-of-mas level against a predicted signal of a few mas.

8. Overlap of the two fields

Matching the two source lists directly, after fitting the offset between the cut-outs (−1.11 / +0.24 pix):

match radius	BLG41 matched	BLG01 matched
0.5 pix	339 / 594 (57%)	352 / 621 (57%)
1.0 pix	546 / 594 (92%)	575 / 621 (93%)
1.5 pix	576 / 594 (97%)	609 / 621 (98%)

These are indeed two observations of nearly the same sky, and the cut-outs span the same pixel range. The unmatched 7% are faint and marginal: median MAG_PSF 17.18 against 16.87 for matched stars, and a detection fraction of 0.54 against 0.84. They are stars one reduction found and the other did not, or blends split differently. Restricted to MAG_PSF < 17 the agreement is 94%.

Only 57% agree within 0.5 pix, so two independent reductions place a common star half a pixel to a pixel apart; 1 pixel is the appropriate radius.

Matching *through* Gaia and OGLE instead gives a misleading 58%, because their identification coverage over our sources is only 90% / 72% (Gaia) and 91% / 77% (OGLE) — sources neither catalogue identifies cannot be paired at all and count as non-overlapping.

9. Verdict

v4 is a clean, gauge-free reduction. Its calibration stars are the best-measured stars produced by this project, and pinning to Gaia is the right way to obtain absolute astrometry. It is **not** adopted as the primary solution, for two reasons:

1. The quantity the project exists to measure — the target's astrometry — comes out 14 to 20% worse, because a 126-star frame is noisier than a 600-star one.
2. The two fields' proper motions agree no better than under the post-hoc tie.

Its most valuable output is not the astrometry but section 7: the direct measurement of how far a blended neighbour is dragged by the event, which applies to the target in every reduction we have made.

10. Figures and files

file	content
report_v4_RMS.png	residual RMS against OGLE I, per axis and field, for this run
report_v4_motion_<field>_target.png	the target, sidereal-month bins, 2x2
report_v4_motion_<field>_cal{1,2,3}.png	three calibration stars, proper motion pinned to Gaia
report_v4_motion_<field>_out{1,2,3}.png	the three l ~ 18 passengers, freely fitted

In every motion figure the columns are the two axes, X left and Y right; the top row keeps the proper motion and the bottom row removes it. **A pinned star's red line is Gaia's proper motion, not a fitted one**, so its lower panels test the frame, whereas the target's and the passengers' test the fit.

Figures

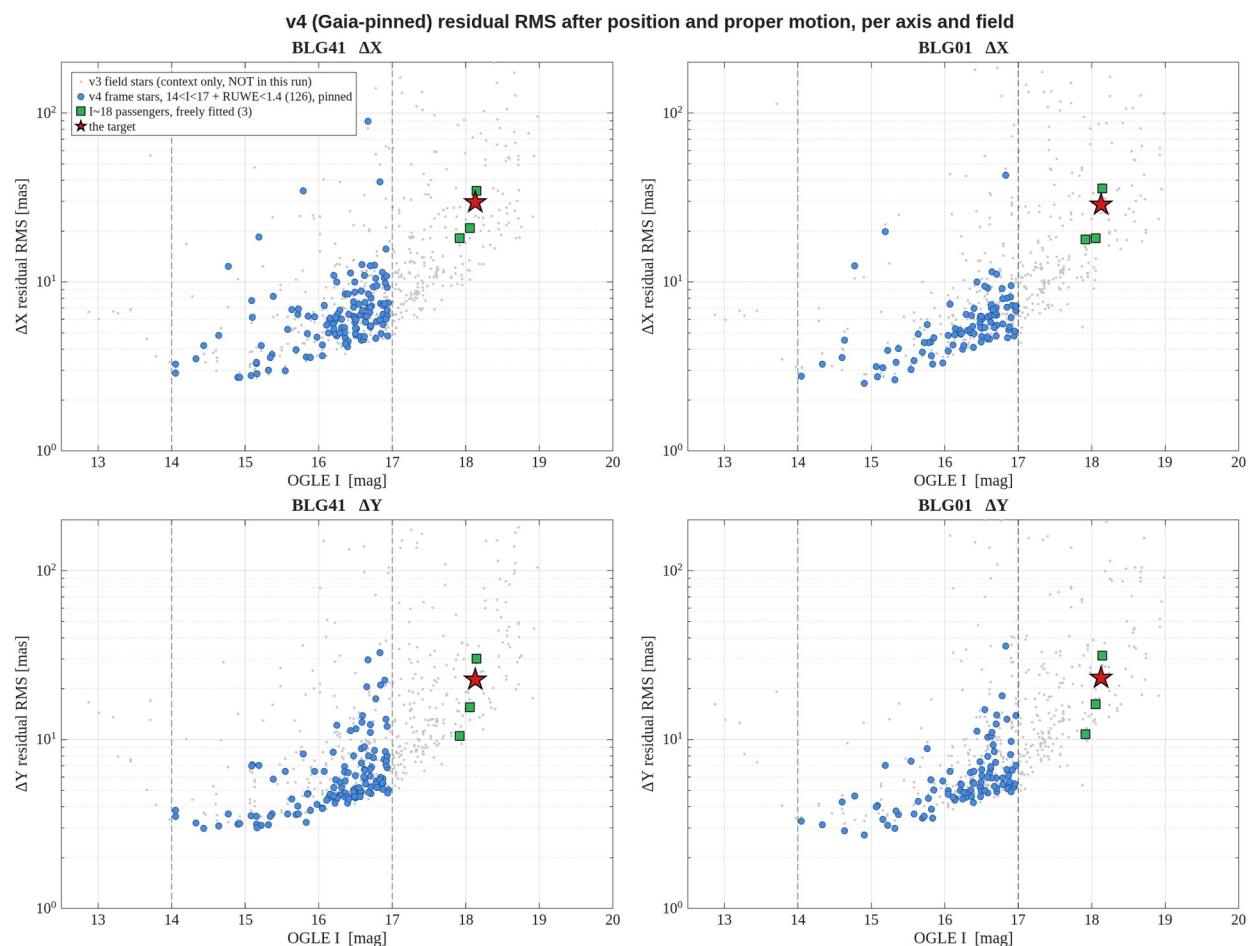


Figure 1. Residual RMS against the OGLE catalogue magnitude, per axis and per field, **for this run**, after position and proper motion have been removed. Blue: the 126 / 96 frame stars, 14<l<17 with RUWE<1.4, pinned to Gaia. Green: the three l~18 passengers, freely fitted. Red: the target. Grey: field stars from the reference reduction, shown for context only — **they are not part of this run**. Dashed lines mark the 14 and 17 mag limits.

report_v4_RMS.png

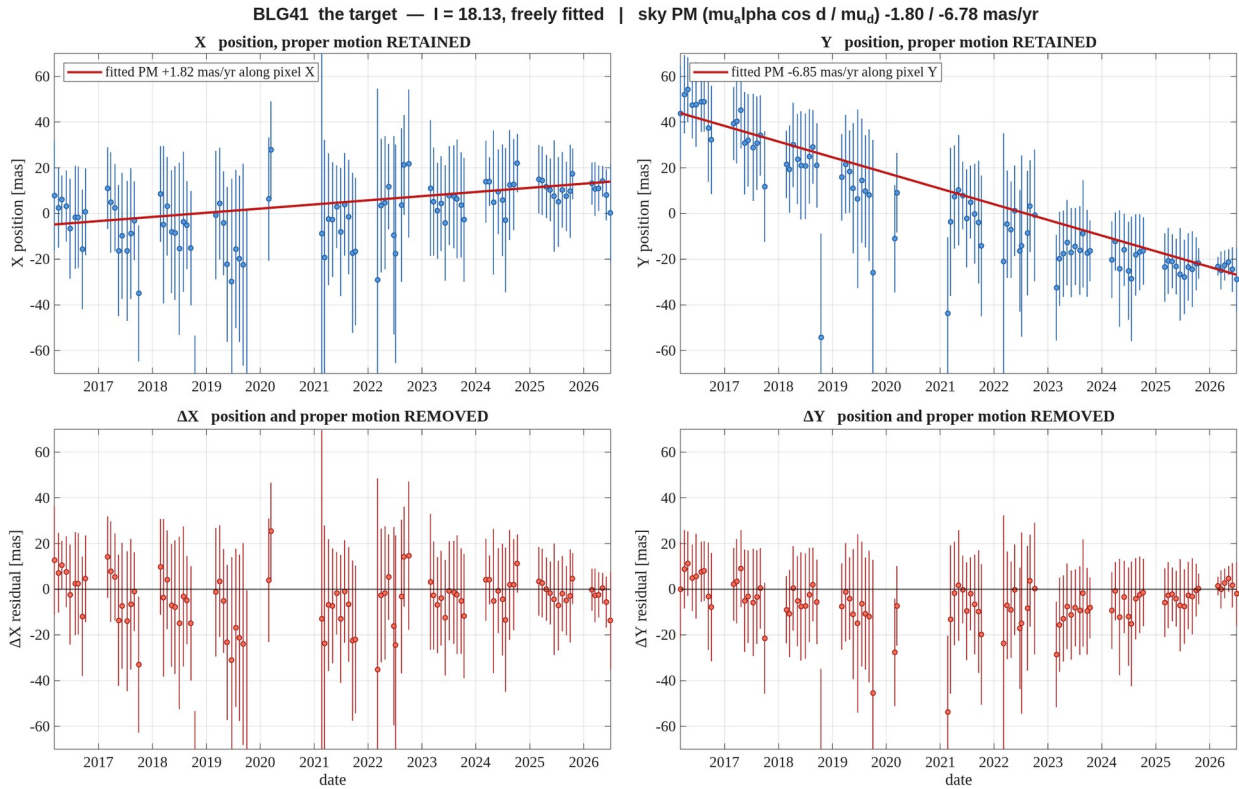


Figure 2. **BLG41** — the target, freely fitted. Binned in one sidereal month; error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the star's magnitude, its distance from the target and its sky proper motion. **A pinned star's red line is Gaia's proper motion, not a fitted one.**
report_v4_motion_BLG41_target.png

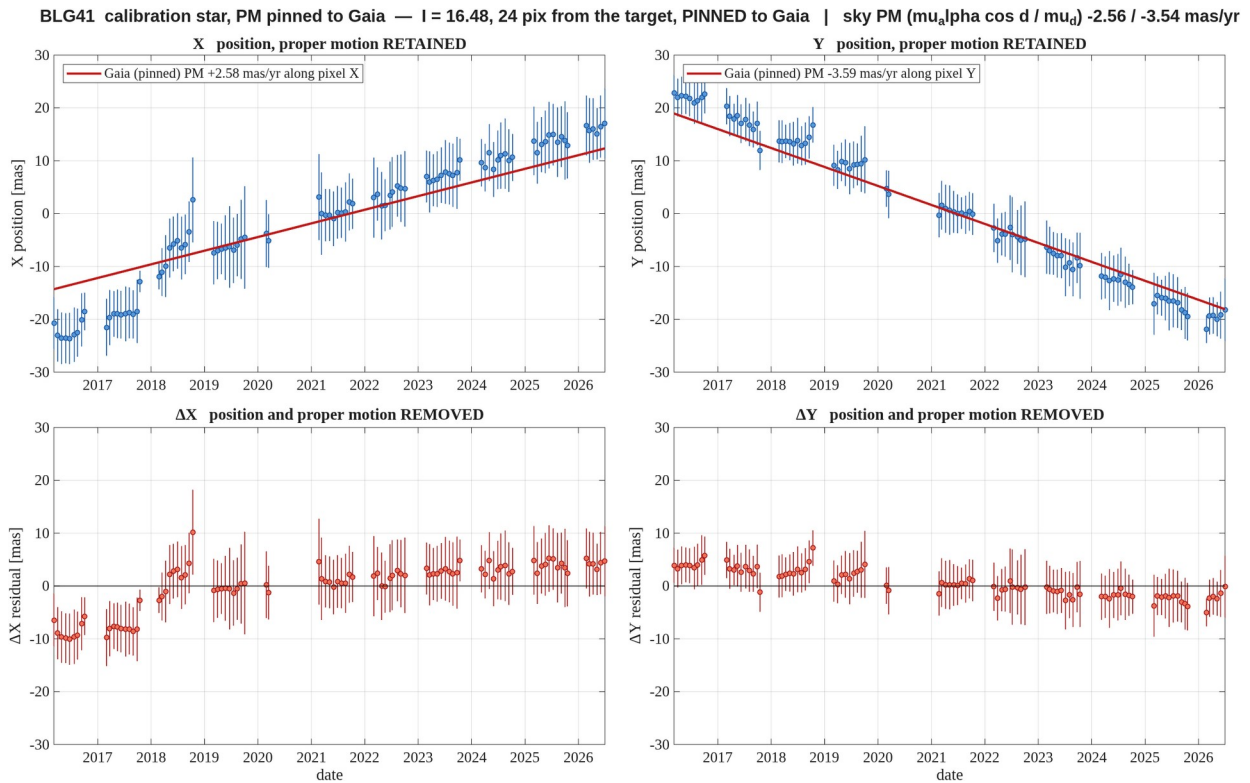


Figure 3. **BLG41** — calibration star, proper motion pinned to Gaia. Binned in one sidereal month; error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the star's magnitude, its distance from the target and its sky proper motion. **A pinned star's red line is Gaia's proper motion, not a fitted one.**
report_v4_motion_BLG41_cal1.png

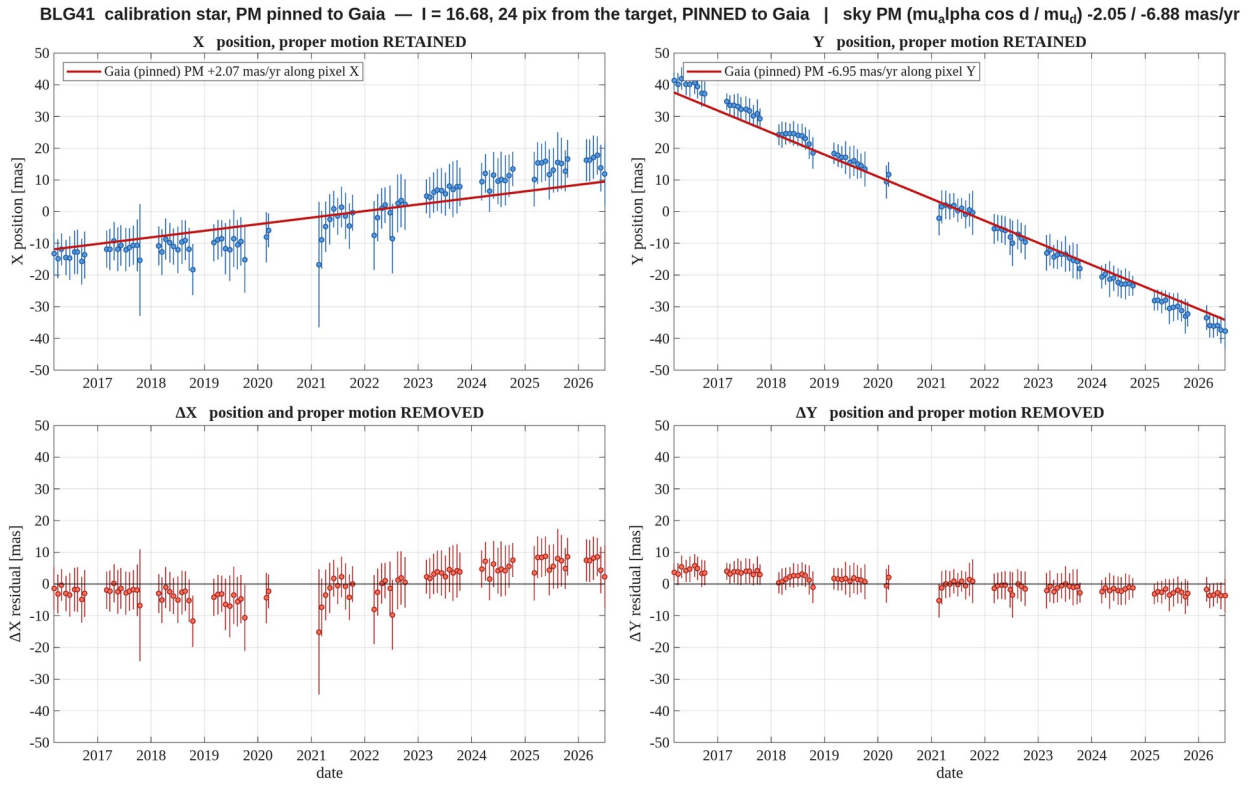


Figure 4. **BLG41** — calibration star, proper motion pinned to Gaia. Binned in one sidereal month; error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the star's magnitude, its distance from the target and its sky proper motion. **A pinned star's red line is Gaia's proper motion, not a fitted one.**

report_v4_motion_BLG41_cal2.png

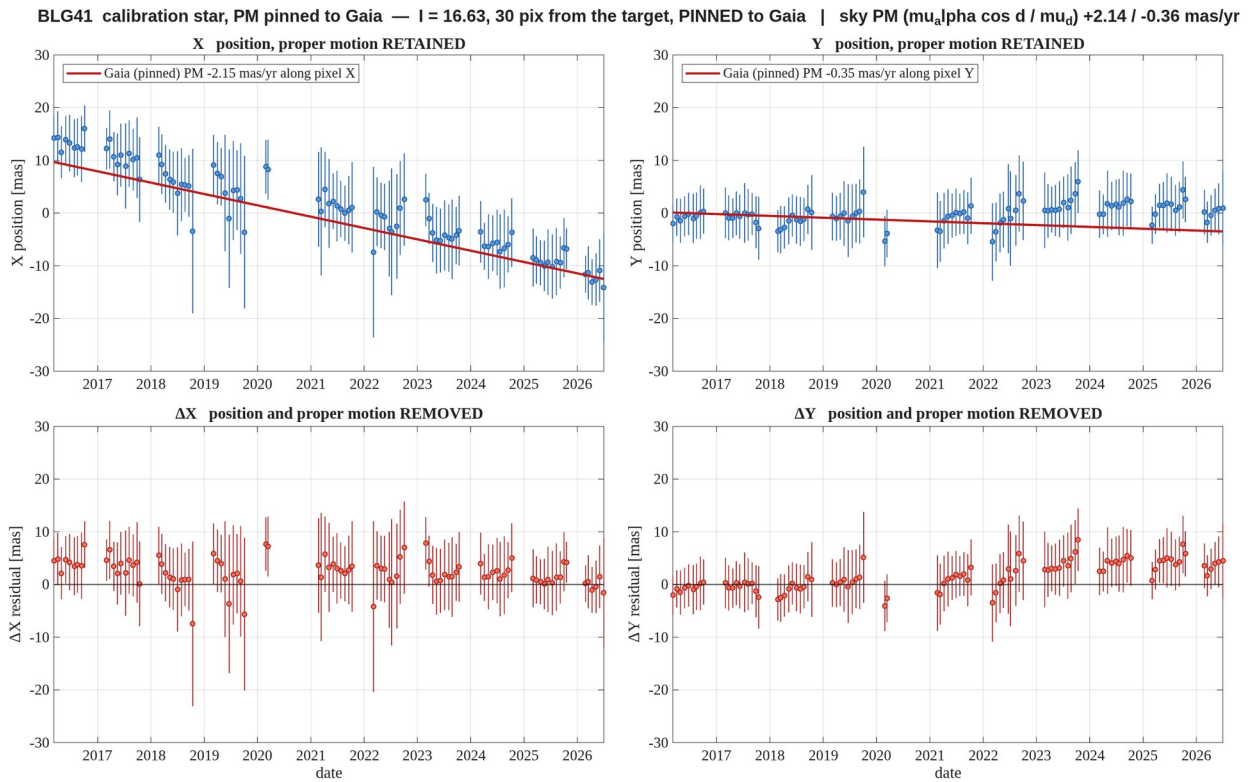


Figure 5. **BLG41** — calibration star, proper motion pinned to Gaia. Binned in one sidereal month; error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the star's magnitude, its distance from the target and its sky proper motion. **A pinned star's red line is Gaia's proper motion, not a fitted one.**

report_v4_motion_BLG41_cal3.png

BLG41 comparison star outside the set, PM fitted — $I = 18.14$, 5 pix from the target, freely fitted | sky PM ($\mu_\alpha \cos d / \mu_\delta$) $-0.63 / -5.47$ mas/yr

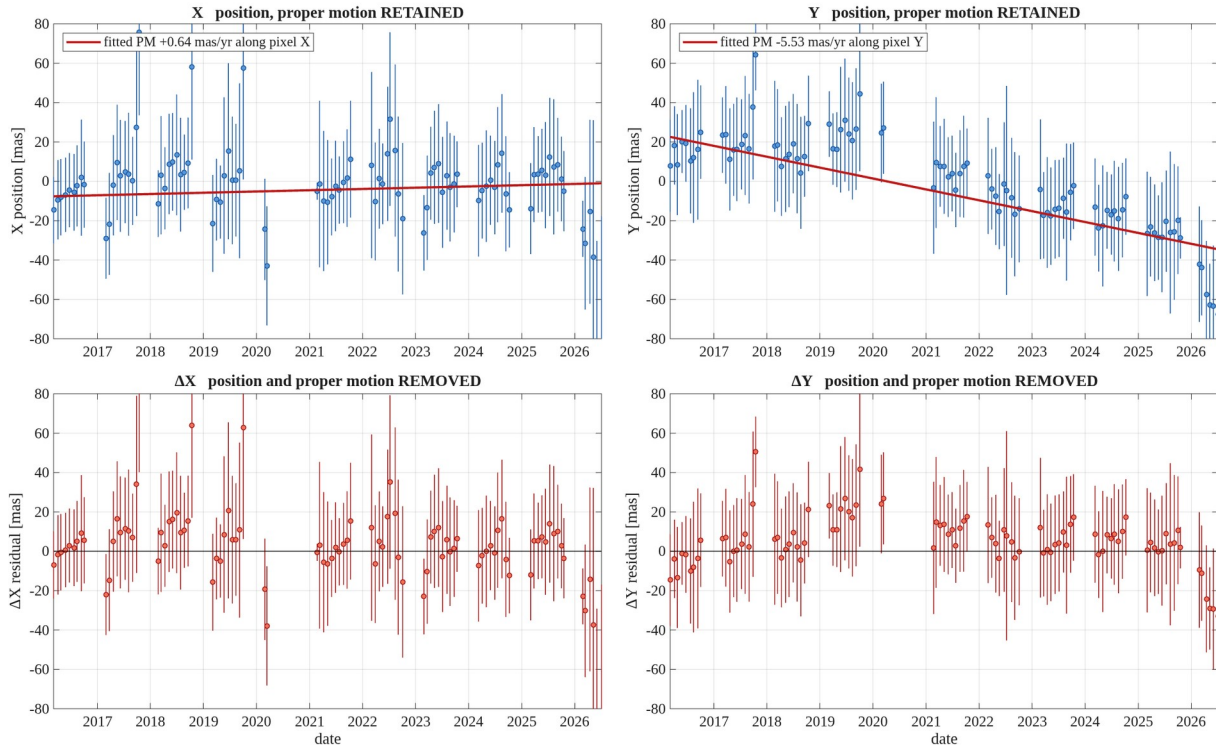


Figure 6. **BLG41** — passenger at the target magnitude, 4.5 pix away — **blended with the target**, see section 7. Binned in one sidereal month; error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the star's magnitude, its distance from the target and its sky proper motion. **A pinned star's red line is Gaia's proper motion, not a fitted one.**

report_v4_motion_BLG41_out1.png

BLG41 comparison star outside the set, PM fitted — $I = 17.91$, 21 pix from the target, freely fitted | sky PM ($\mu_\alpha \cos d / \mu_\delta$) $-6.81 / -9.99$ mas/yr

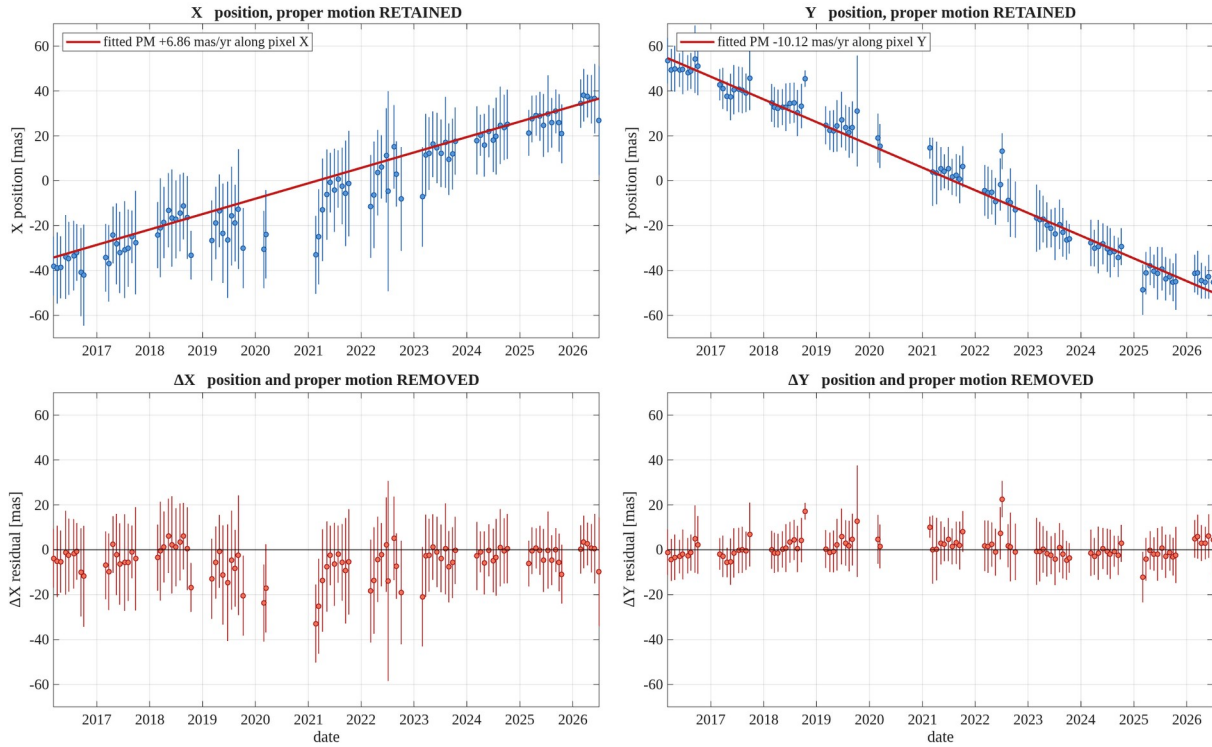


Figure 7. **BLG41** — passenger at the target magnitude, 21 pix away, freely fitted. Binned in one sidereal month; error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the star's magnitude, its distance from the target and its sky proper motion. **A pinned star's red line is Gaia's proper motion, not a fitted one.**

report_v4_motion_BLG41_out2.png

BLG41 comparison star outside the set, PM fitted — $I = 18.05$, 48 pix from the target, freely fitted | sky PM ($\mu_\alpha \cos d / \mu_d$) $+3.06 / -6.34$ mas/yr

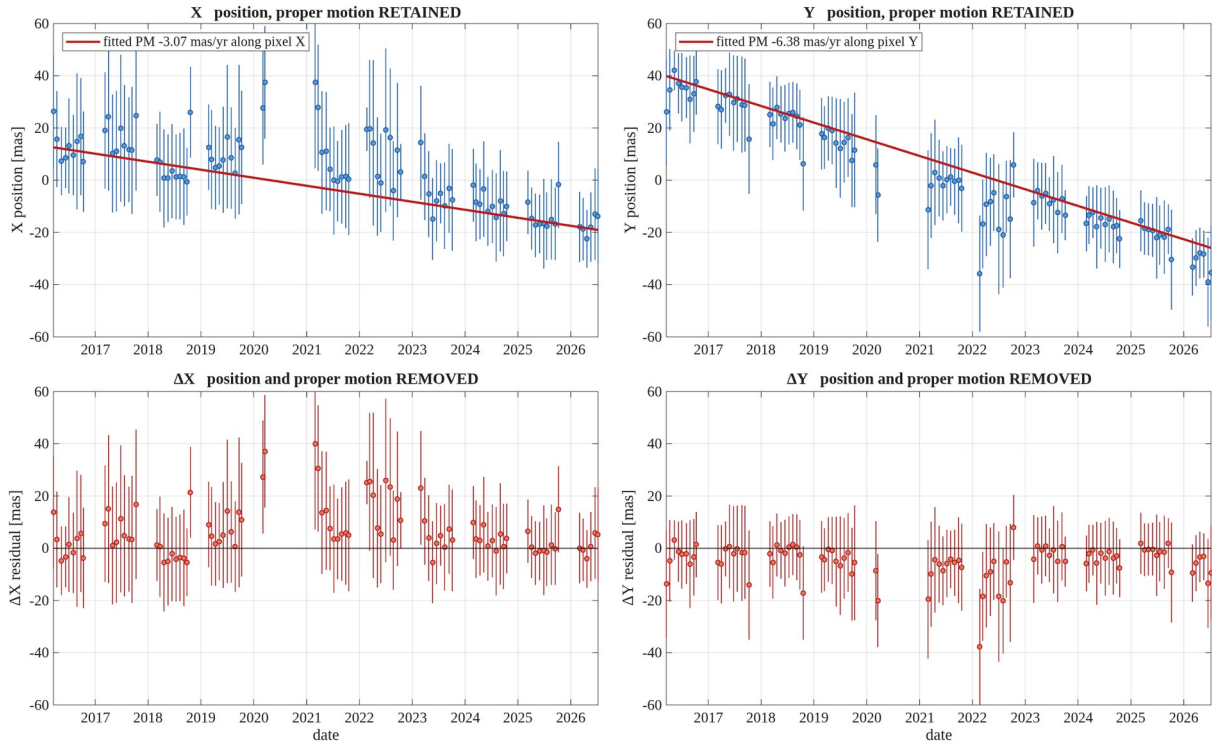


Figure 8. **BLG41** — passenger at the target magnitude, 48 pix away, freely fitted. Binned in one sidereal month; error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the star's magnitude, its distance from the target and its sky proper motion. **A pinned star's red line is Gaia's proper motion, not a fitted one.**

report_v4_motion_BLG41_out3.png

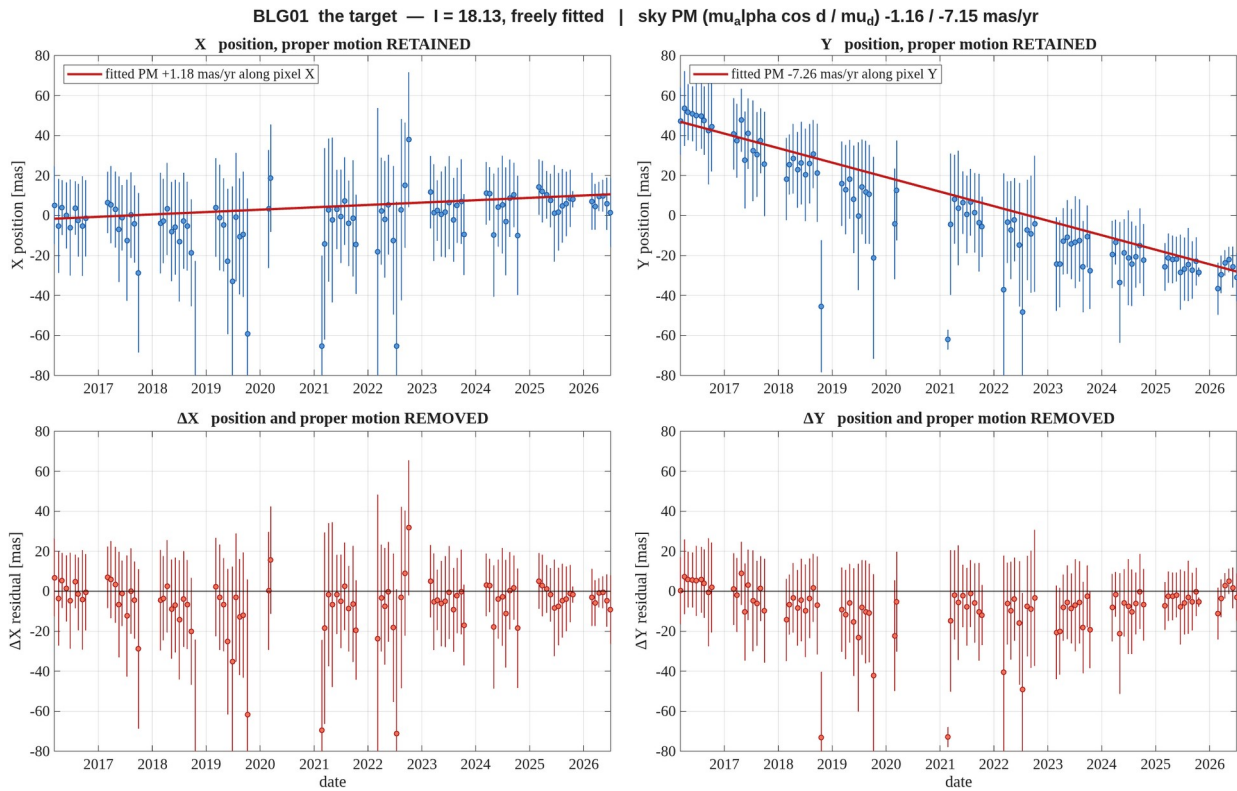


Figure 9. **BLG01** — the target, freely fitted. Binned in one sidereal month; error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the star's magnitude, its distance from the target and its sky proper motion. **A pinned star's red line is Gaia's proper motion, not a fitted one.**

report_v4_motion_BLG01_target.png

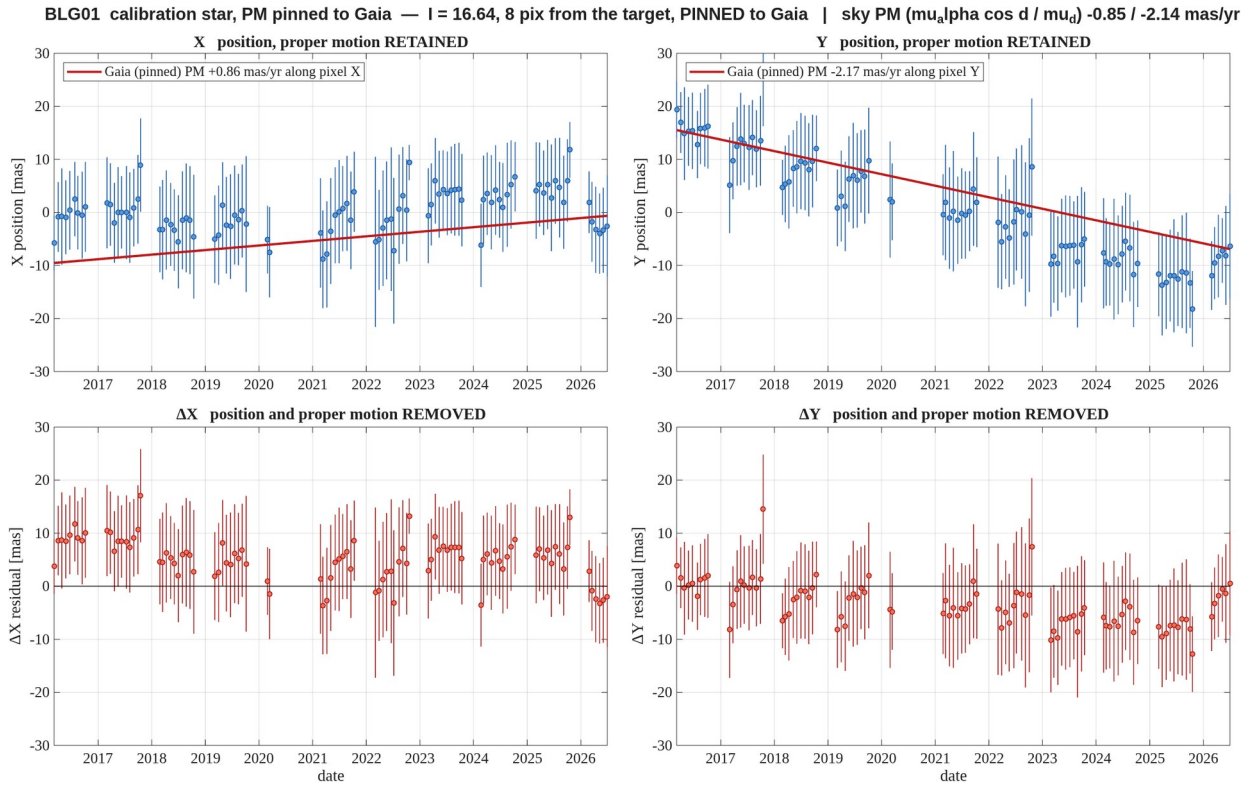


Figure 10. **BLG01** — calibration star, proper motion pinned to Gaia. Binned in one sidereal month; error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the star's magnitude, its distance from the target and its sky proper motion. **A pinned star's red line is Gaia's proper motion, not a fitted one.**

report_v4_motion_BLG01_cal1.png

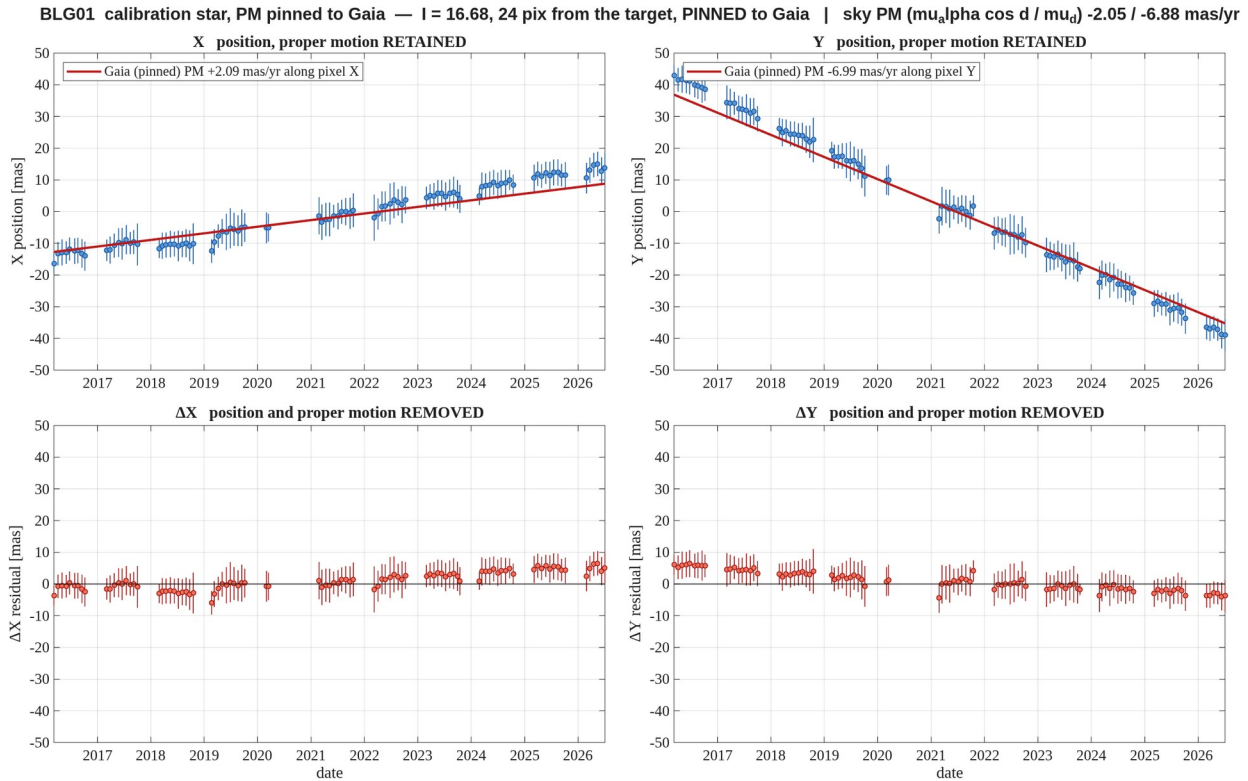


Figure 11. **BLG01** — calibration star, proper motion pinned to Gaia. Binned in one sidereal month; error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the star's magnitude, its distance from the target and its sky proper motion. **A pinned star's red line is Gaia's proper motion, not a fitted one.**

report_v4_motion_BLG01_cal2.png

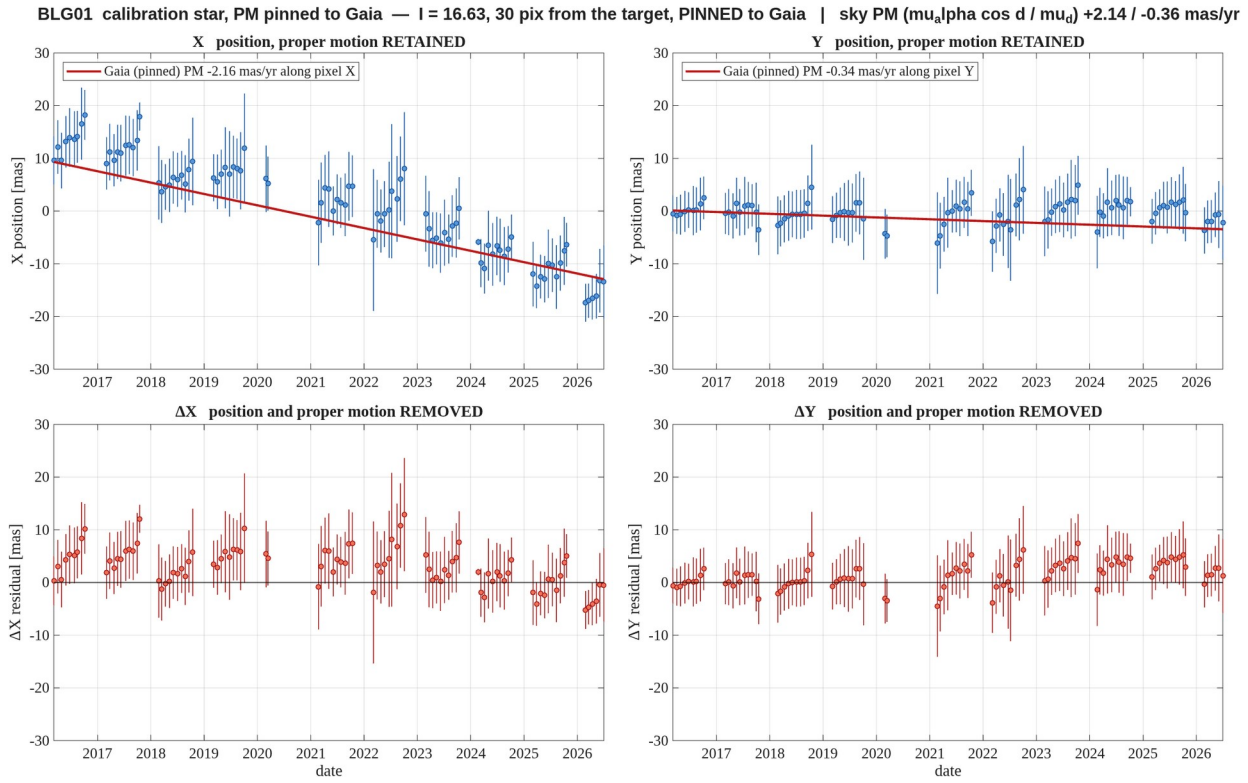


Figure 12. **BLG01** — calibration star, proper motion pinned to Gaia. Binned in one sidereal month; error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the star's magnitude, its distance from the target and its sky proper motion. **A pinned star's red line is Gaia's proper motion, not a fitted one.**

report_v4_motion_BLG01_cal3.png

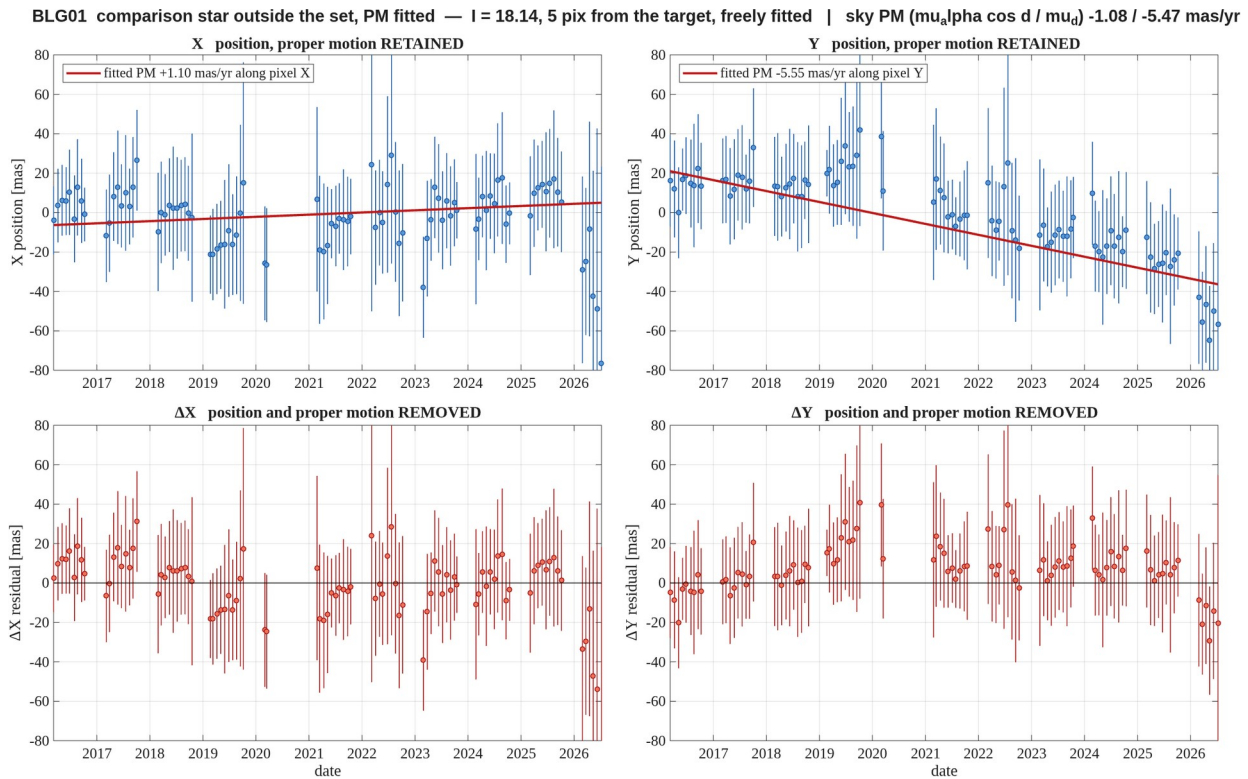


Figure 13. **BLG01** — passenger at the target magnitude, 4.5 pix away — **blended with the target**, see section 7. Binned in one sidereal month; error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the star's magnitude, its distance from the target and its sky proper motion. **A pinned star's red line is Gaia's proper motion, not a fitted one.**

report_v4_motion_BLG01_out1.png

BLG01 comparison star outside the set, PM fitted — $I = 17.91$, 21 pix from the target, freely fitted | sky PM ($\mu_\alpha \cos d / \mu_\delta$) $-6.94 / -10.06$ mas/yr

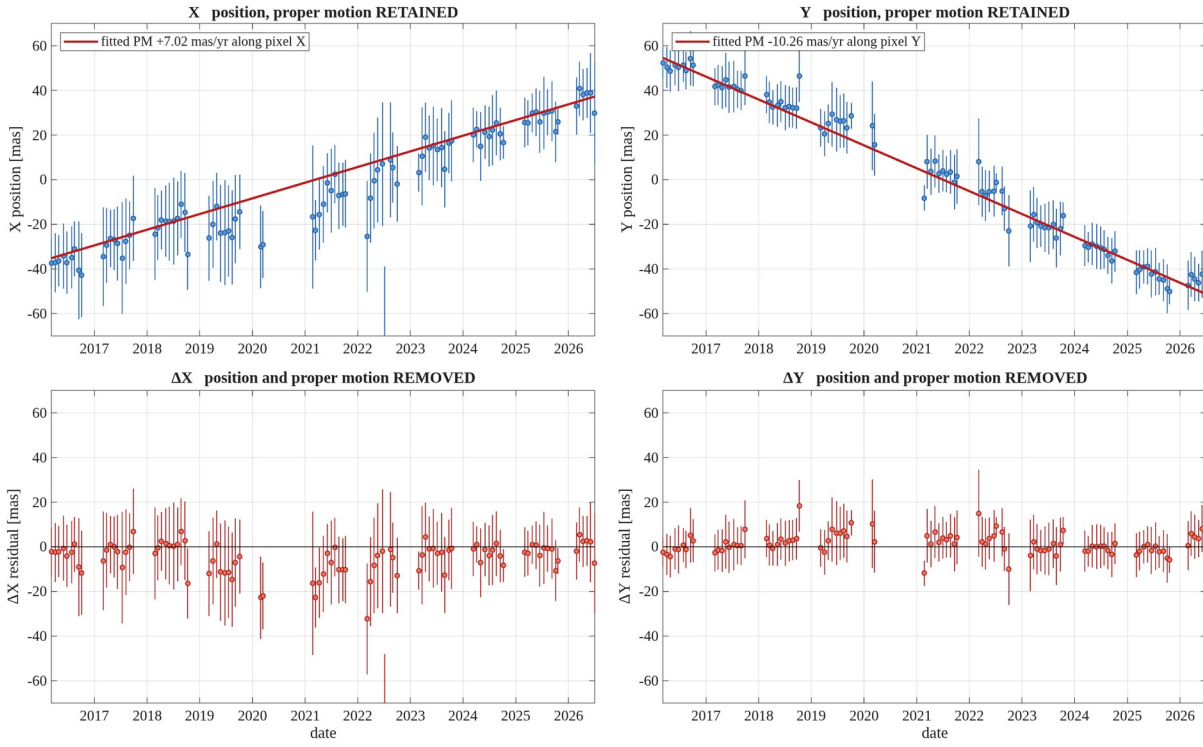


Figure 14. **BLG01** — passenger at the target magnitude, 21 pix away, freely fitted. Binned in one sidereal month; error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the star's magnitude, its distance from the target and its sky proper motion. **A pinned star's red line is Gaia's proper motion, not a fitted one.**

report_v4_motion_BLG01_out2.png

BLG01 comparison star outside the set, PM fitted — $I = 18.05$, 49 pix from the target, freely fitted | sky PM ($\mu_\alpha \cos d / \mu_\delta$) $+2.80 / -6.48$ mas/yr

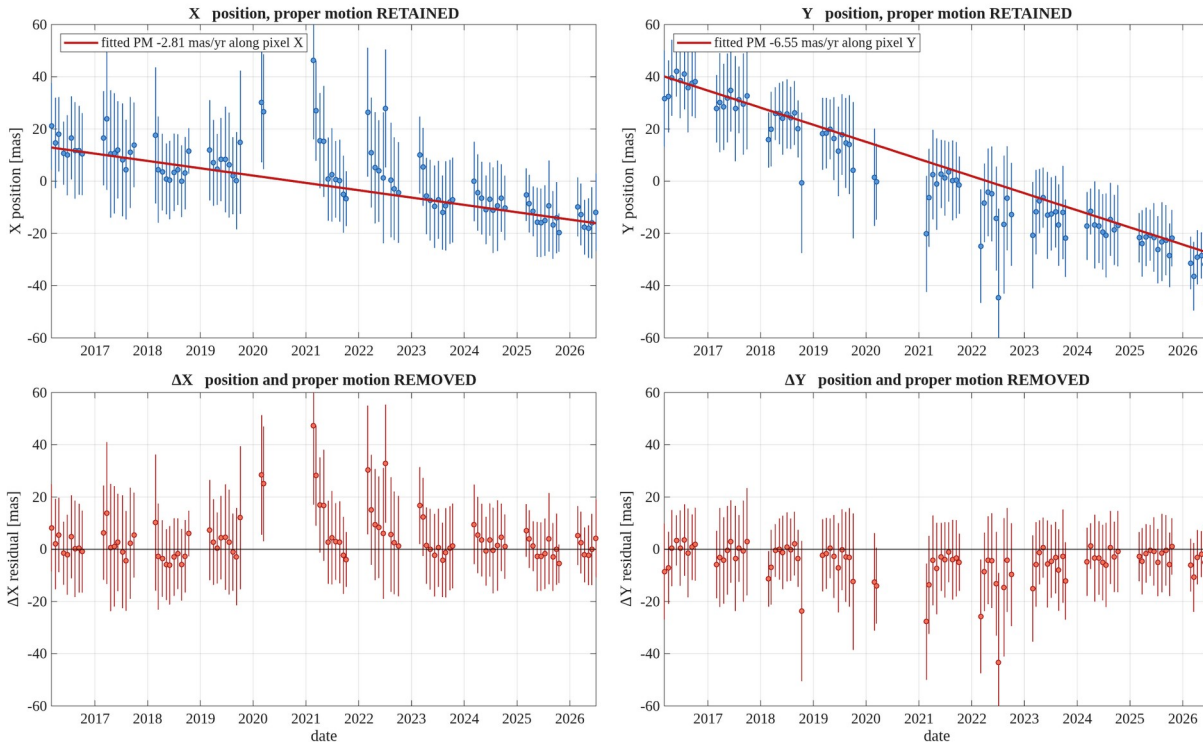


Figure 15. **BLG01** — passenger at the target magnitude, 48 pix away, freely fitted. Binned in one sidereal month; error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the star's magnitude, its distance from the target and its sky proper motion. **A pinned star's red line is Gaia's proper motion, not a fitted one.**

report_v4_motion_BLG01_out3.png